1. spendingAnalyzer.js

Purpose

Raw spending statistics.

It should answer:

- How much spent?
- How many transactions?
- Daily average?
- Weekly average?
- Monthly average?
- Largest expense?
- Smallest expense?
- Average transaction amount?

It should NOT care about budgets or habits.

Returns

```
{
  totalSpent,
  transactionCount,
  averageTransaction,
  dailyAverage,
  weeklyAverage,
  monthlyAverage,
  largestExpense,
  smallestExpense
}
```

---

2. budgetAnalyzer.js

This is ONLY about budget.

Questions:

- Budget utilized %
- Remaining budget
- Overspending?
- Budget left

- Budget streak
- Days until exhaustion
- Projected overspend

Returns

```
{
  budget,
  spent,
  remaining,
  utilization,
  status,
  projectedEndOfMonth,
  streak
}
```

---

3. categoryAnalyzer.js

Anything category-related.

Questions

- Top category
- Least category
- Category percentages
- Spending distribution
- Biggest category increase
- Biggest category decrease

Returns

```
{
  topCategory,
  leastCategory,
  categoryDistribution,
  categoryGrowth,
  biggestJump,
  biggestDrop
}
```

---

4. trendAnalyzer.js

This compares time periods.

Questions

- Compared to yesterday
- Compared to last week
- Compared to last month
- Compared to previous quarter

Returns

```
{
  dailyTrend,
  weeklyTrend,
  monthlyTrend,
  yearlyTrend,
  spendingDirection
}
```

---

5. habitAnalyzer.js

This is behavioral analysis.

Not financial health.

Questions

- Weekend vs weekday
- Night spending
- Morning spending
- Micro spending
- Impulse spending
- Subscription pattern
- Shopping frequency

Returns

```
{
  weekdayAverage,
  weekendAverage,
  microSpending,
```

```
    impulseScore,
    frequentCategories,
    shoppingPattern
}
```

---

6. healthAnalyzer.js

This is the highest-level analyzer.

It DOES NOT calculate raw statistics.

Instead, it consumes outputs from all other analyzers.

Example

```
Budget Analyzer
↓
90% budget used
Trend Analyzer
↓
+35%
Habit Analyzer
↓
Impulse score high
Health Analyzer
↓
Financial Health = 42/100
```

This analyzer combines everything.

Returns

```
{
    stabilityScore,
    healthScore,
    riskLevel,
    strengths,
    weaknesses,
    recommendations
}
```

---

7. reportGenerator.js

This should never perform analysis.

Its job is only to merge outputs.

```
{
  spending: ...,
  budget: ...,
  category: ...,
  trend: ...,
  habit: ...,
  health: ...
}
```

and maybe

```
{
  generatedAt,
  month,
  report
}
```

It acts as the assembler for your frontend or SIA.

analyticsEngine.generateReport()

↓

fetchTransactions()

fetchBudgets()

↓

spendingAnalyzer.analyze(transactions)

budgetAnalyzer.analyze({

budgets,

transactions

})

categoryAnalyzer.analyze(transactions)

trendAnalyzer.analyze(transactions)

habitAnalyzer.analyze(transactions)

healthAnalyzer.analyze({

spending,

budget,

trend,

habit,

category

})